

Bath Correlation Function and Spectral Density for Harmonic Oscillator Baths

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1 Harmonic Oscillator Bath

In this note, we will derive the analytical expression of the bath correlation function (or reservoir correlation function) for harmonic oscillator baths. We assume that the bath consists of a set of uncoupled harmonic oscillators. The system-bath coupling results in displacements of the equilibrium positions of these oscillators, but does not change the vibrational frequencies or the normal mode coordinates.

Let $\hat{q}_j$ and $\hat{p}_j$ denote the operators for the normal mode coordinate and the conjugate momentum of mode j , the Hamiltonian of the model system can be written as

$$\hat{H} = \sum_m \left[\varepsilon_m + \sum_j \left(\frac{1}{2} \hat{p}_j^2 + \frac{1}{2} \omega_j^2 (\hat{q}_j - \delta q_{j,m})^2 \right) \right] |m\rangle \langle m| + \sum_{m \neq n} V_{mn} |m\rangle \langle n| \quad (1)$$

where ε_m is the energy of state $|m\rangle$ which is independent of the bath, and V_{mn} is the coupling between states $|m\rangle$ and $|n\rangle$. $\delta q_{j,m}$ is the displacement of the equilibrium position of along normal mode j on state $|m\rangle$. This Hamiltonian can be rewritten as

$$\begin{aligned} \hat{H} &= \hat{H}_S + \hat{H}_B + \hat{H}_{SB} \\ &= \sum_m \left(\varepsilon_m + \sum_j \frac{1}{2} \omega_j^2 \delta q_{j,m}^2 \right) |m\rangle \langle m| + \sum_{m \neq n} V_{mn} |m\rangle \langle n| \\ &\quad + \sum_j \left(\frac{1}{2} \hat{p}_j^2 + \frac{1}{2} \omega_j^2 \hat{q}_j^2 \right) \\ &\quad - \sum_m \sum_j \omega_j^2 \delta q_{j,m} \hat{q}_j |m\rangle \langle m| \end{aligned} \quad (2)$$

Here, the first line is the Hamiltonian for the reactive system $\hat{H}_S$, the second line is the bath Hamiltonian $\hat{H}_B$, and the third line is the system-bath coupling term $\hat{H}_{SB}$. We have used the identity $\sum_m |m\rangle \langle m| = 1$.

Next, we introduce the following quantities: the **reorganization energy**

$$\lambda_m = \sum_j \frac{1}{2} \omega_j^2 \delta q_{j,m}^2 = \sum_j \hbar \omega_j X_{j,m} \quad (3)$$

the **Huang-Rhys factor**

$$X_{j,m} = \frac{\omega_j}{2\hbar} \delta q_{j,m}^2 \quad (4)$$

and the (frequency-dependent) **coupling strength**

$$\gamma_{j,m} = \hbar\omega_j \sqrt{X_{j,m}} = \sqrt{\frac{\hbar\omega_j}{2}} \omega_j \delta q_{j,m} \quad (5)$$

We also express the operators $\hat{q}_j$ and $\hat{p}_j$ in terms of the creation and annihilation operators $\hat{b}_j^\dagger$ and $\hat{b}_j$ for bath mode j ,

$$\hat{q}_j = \sqrt{\frac{\hbar}{2\omega_j}} (\hat{b}_j^\dagger + \hat{b}_j), \quad \hat{p}_j = i\sqrt{\frac{\hbar\omega_j}{2}} (\hat{b}_j^\dagger - \hat{b}_j) \quad (6)$$

The model Hamiltonian then becomes

$$\begin{aligned} \hat{H} = & \sum_m (\varepsilon_m + \lambda_m) |m\rangle\langle m| + \sum_{m \neq n} V_{mn} |m\rangle\langle n| \\ & + \sum_j \hbar\omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right) \\ & - \sum_m \sum_j \gamma_{j,m} (\hat{b}_j^\dagger + \hat{b}_j) |m\rangle\langle m| \end{aligned} \quad (7)$$

2 Bath Correlation Function and Spectral Density

In the theory of open-quantum systems, a central quantity that governs the dissipative dynamics of the reactive system is the bath correlation function. Assuming the system-bath coupling has the form

$$\hat{H}_{\text{SB}} = \sum_m \hat{A}_m \otimes \hat{B}_m \quad (8)$$

where $\hat{A}_m$ is an operator only act on the reactive system and $\hat{B}_m$ is an operator only act on the bath. Then the bath correlation function is defined as*

$$C_{mn}(t) = \text{Tr}_{\text{B}} \left(\hat{B}_m(t) \hat{B}_n(0) \hat{\rho}_{\text{B}} \right) \quad (9)$$

Here,

$$\hat{B}_m(t) = e^{\frac{i}{\hbar} \hat{H}_{\text{B}} t} \hat{B}_m e^{-\frac{i}{\hbar} \hat{H}_{\text{B}} t} \quad (10)$$

$$\hat{\rho}_{\text{B}} = \frac{e^{-\beta \hat{H}_{\text{B}}}}{\text{Tr}_{\text{B}} e^{-\beta \hat{H}_{\text{B}}}} \equiv \frac{1}{Z_{\text{B}}} e^{-\beta \hat{H}_{\text{B}}} \quad (11)$$

$\hat{H}_{\text{B}}$ is the bath Hamiltonian, $\beta = 1/k_{\text{B}}T$, and Z_{B} is the partition function of the bath. In this note, we will only focus the case where $m = n$. In this case, $C_{mm}(t)$ will be abbreviated as $C_m(t)$.

For the Hamiltonian defined in Eq. (7), we have $\hat{A}_m = |m\rangle\langle m|$ and

$$\hat{B}_m = - \sum_j \gamma_{j,m} (\hat{b}_j^\dagger + \hat{b}_j) \quad (12)$$

*The bath correlation function can be defined in different ways. In some literature, it has a $1/\hbar^2$ prefactor.

In the subsequent derivation, we will omit the subscript m for notational simplicity. Substitute the expression into the definition of the bath correlation function, we have

$$C(t) = \sum_{j,j'} \gamma_j \gamma_{j'} \text{Tr}_B \left[e^{\frac{i}{\hbar} \hat{H}_B t} (\hat{b}_j^\dagger + \hat{b}_j) e^{-\frac{i}{\hbar} \hat{H}_B t} (\hat{b}_{j'}^\dagger + \hat{b}_{j'}) \hat{\rho}_B \right] \quad (13)$$

Let $|M\rangle = \bigotimes_j |M_j\rangle$ denotes the eigenstate of the bath Hamiltonian with M_j being the oscillator quantum number. The trace can be calculated as

$$\begin{aligned} C(t) &= \sum_{j,j'} \gamma_j \gamma_{j'} \sum_{M,N} \langle M | e^{\frac{i}{\hbar} \hat{H}_B t} (\hat{b}_j^\dagger + \hat{b}_j) e^{-\frac{i}{\hbar} \hat{H}_B t} | N \rangle \langle N | (\hat{b}_{j'}^\dagger + \hat{b}_{j'}) \hat{\rho}_B | M \rangle \\ &= \sum_{j,j'} \gamma_j \gamma_{j'} \sum_{M,N} f(E_M) \langle M | (\hat{b}_j^\dagger + \hat{b}_j) | N \rangle \langle N | (\hat{b}_{j'}^\dagger + \hat{b}_{j'}) | M \rangle e^{i\omega_{MN}t} \end{aligned} \quad (14)$$

where E_M and E_N are the energies of the bath states $|M\rangle$ and $|N\rangle$, $\omega_{MN} = (E_M - E_N)/\hbar$, and $f(E_M)$ is the Boltzmann distribution of the bath energy.

It's more convenient to calculate the Fourier transform of the bath correlation function

$$\begin{aligned} C(\omega) &= \int_{-\infty}^{\infty} dt e^{i\omega t} C(t) \\ &= 2\pi \sum_{j,j'} \gamma_j \gamma_{j'} \sum_{M,N} f(E_M) \langle M | (\hat{b}_j^\dagger + \hat{b}_j) | N \rangle \langle N | (\hat{b}_{j'}^\dagger + \hat{b}_{j'}) | M \rangle \delta(\omega + \omega_{MN}) \end{aligned} \quad (15)$$

For the states $|M\rangle = \bigotimes_j |M_j\rangle$ and $|N\rangle = \bigotimes_j |N_j\rangle$, the matrix element is

$$\begin{aligned} \langle M | (\hat{b}_j^\dagger + \hat{b}_j) | N \rangle &= \langle M_j | (\hat{b}_j^\dagger + \hat{b}_j) | N_j \rangle \prod_{k \neq j} \langle M_k | N_k \rangle \\ &= \left(\sqrt{N_j} \delta_{M_j, N_j-1} + \sqrt{N_j+1} \delta_{M_j, N_j+1} \right) \prod_{k \neq j} \delta_{M_k, N_k} \end{aligned} \quad (16)$$

$$\langle N | (\hat{b}_{j'}^\dagger + \hat{b}_{j'}) | M \rangle = \left(\sqrt{M_{j'}} \delta_{N_{j'}, M_{j'}-1} + \sqrt{M_{j'}+1} \delta_{N_{j'}, M_{j'}+1} \right) \prod_{k' \neq j'} \delta_{M_{k'}, N_{k'}} \quad (17)$$

When $j = j'$,

$$\begin{aligned} &\langle M | (\hat{b}_j^\dagger + \hat{b}_j) | N \rangle \langle N | (\hat{b}_j^\dagger + \hat{b}_j) | M \rangle \\ &= \left(\sqrt{N_j} \delta_{M_j, N_j-1} + \sqrt{N_j+1} \delta_{M_j, N_j+1} \right) \left(\sqrt{M_j} \delta_{N_j, M_j-1} + \sqrt{M_j+1} \delta_{N_j, M_j+1} \right) \\ &\quad \times \prod_{k \neq j} \delta_{M_k, N_k} \prod_{k' \neq j} \delta_{M_{k'}, N_{k'}} \\ &= \left[(M_j + 1) \delta_{N_j, M_j+1} + M_j \delta_{N_j, M_j-1} \right] \prod_{k \neq j} \delta_{M_k, N_k} \prod_{k' \neq j} \delta_{M_{k'}, N_{k'}} \\ &= \left[(M_j + 1) \delta_{N_j, M_j+1} + M_j \delta_{N_j, M_j-1} \right] \prod_{k \neq j} \delta_{M_k, N_k} \end{aligned} \quad (18)$$

The last equality holds because $\delta_{ab} \delta_{ab} = \delta_{ab}$. Since $f(E_M) = \prod_l f(E_{M_l})$, $\sum_{M_l} f(E_{M_l}) = 1$, and

$\delta(\omega + \omega_{MN}) = \delta(\omega + \sum_l \omega_{M_l N_l})$, we have

$$\begin{aligned}
& \sum_{M,N} f(E_M) [(M_j + 1)\delta_{N_j, M_j+1} + M_j\delta_{N_j, M_j-1}] \prod_{k \neq j} \delta_{M_k, N_k} \delta(\omega + \omega_{MN}) \\
&= \sum_{M_j, N_j} f(E_{M_j}) [(M_j + 1)\delta_{N_j, M_j+1} + M_j\delta_{N_j, M_j-1}] \delta(\omega + \omega_{M_j N_j}) \\
&= \sum_{M_j} f(E_{M_j}) [(M_j + 1)\delta(\omega - \omega_j) + M_j\delta(\omega + \omega_j)] \\
&= (n(\omega_j) + 1)\delta(\omega - \omega_j) + n(\omega_j)\delta(\omega + \omega_j)
\end{aligned} \tag{19}$$

where

$$n(\omega_j) = \sum_{M_j} f(E_{M_j}) M_j = \frac{1}{e^{\beta \hbar \omega_j} - 1} \tag{20}$$

is the mean occupation number of a harmonic oscillator mode.

When $j \neq j'$,

$$\begin{aligned}
& \langle M | (\hat{b}_j^\dagger + \hat{b}_j) | N \rangle \langle N | (\hat{b}_{j'}^\dagger + \hat{b}_{j'}) | M \rangle \\
&= \left(\sqrt{N_j} \delta_{M_j, N_j-1} + \sqrt{N_j + 1} \delta_{M_j, N_j+1} \right) \left(\sqrt{M_{j'}} \delta_{N_{j'}, M_{j'}-1} + \sqrt{M_{j'} + 1} \delta_{N_{j'}, M_{j'}+1} \right) \\
&\quad \times \prod_{k \neq j} \delta_{M_k, N_k} \prod_{k' \neq j'} \delta_{M_{k'}, N_{k'}} \\
&= \left(\sqrt{N_j} \delta_{M_j, N_j-1} + \sqrt{N_j + 1} \delta_{M_j, N_j+1} \right) \left(\sqrt{M_{j'}} \delta_{N_{j'}, M_{j'}-1} + \sqrt{M_{j'} + 1} \delta_{N_{j'}, M_{j'}+1} \right) \\
&\quad \times \delta_{M_j, N_j} \delta_{M_{j'}, N_{j'}} \prod_{k \neq j, k \neq j'} \delta_{M_k, N_k} \\
&= 0
\end{aligned} \tag{21}$$

Combine the two cases together, we derived the **Fourier transform of the bath correlation function for harmonic bath**,

$$C(\omega) = 2\pi \sum_j \gamma_j^2 [(n(\omega_j) + 1)\delta(\omega - \omega_j) + n(\omega_j)\delta(\omega + \omega_j)] \tag{22}$$

Since $n(\omega_j)\delta(\omega - \omega_j) = n(\omega)\delta(\omega - \omega_j)$ and $n(\omega_j)\delta(\omega + \omega_j) = n(-\omega)\delta(\omega + \omega_j) = -(n(\omega) + 1)\delta(\omega + \omega_j)$, $C(\omega)$ can be written as

$$C(\omega) = 2\pi \sum_j \gamma_j^2 [(1 + n(\omega))][\delta(\omega - \omega_j) - \delta(\omega + \omega_j)] \tag{23}$$

The **spectral density** is defined as[†]

$$J(\omega) = \sum_j \gamma_j^2 \delta(\omega - \omega_j) \tag{24}$$

[†]In the literature, there are multiple ways to define the spectral density. These definitions differ from each other by a prefactor containing π , $\hbar$, and/or ω . For our definition here, $J(\omega)$ has the dimension of $[E]^2[t]$.

Using this definition,

$$C(\omega) = 2\pi[(1 + n(\omega))][J(\omega) - J(-\omega)] \quad (25)$$

Performing the inverse Fourier transform and using the relations $1 + 2n(\omega) = \coth \frac{\beta\hbar\omega}{2}$ and $J(\omega) = 0$ when $\omega < 0$, we obtain the **bath correlation function for harmonic oscillator bath**,

$$C(t) = \int_0^\infty d\omega J(\omega) \left[\cos(\omega t) \coth \frac{\beta\hbar\omega}{2} - i \sin(\omega t) \right] \quad (26)$$

Note that the spectral density is temperature-independent, but the bath correlation function depends on the temperature.

3 Classical and Semiclassical Bath Correlation Function

For a harmonic oscillator bath, the bath correlation function $C(t)$ can be calculated using Eq. (26) as long as the spectral density $J(\omega)$ is known. However, for large systems, calculating $J(\omega)$ using its definition will be challenging. A practical way to obtain the spectral density is to perform classical molecular dynamics simulations and extract $J(\omega)$ from a classical bath correlation function.

Let $B(\mathbf{q}, \mathbf{p})$ be the classical quantity corresponding to the bath operator $\hat{B}$, $H_B(\mathbf{q}, \mathbf{p})$ be the classical bath Hamiltonian, and $f(\mathbf{q}, \mathbf{p})$ be the distribution function for the phase space variables $\mathbf{q}$ and $\mathbf{p}$. The classical bath correlation function is defined as

$$C_{\text{cl}}(t) = \iint d\mathbf{q}_0 d\mathbf{p}_0 f(\mathbf{q}_0, \mathbf{p}_0) B(\mathbf{q}(t), \mathbf{p}(t)) B(\mathbf{q}_0, \mathbf{p}_0) \quad (27)$$

where $(\mathbf{q}(t), \mathbf{p}(t))$ represent a classical trajectory in phase space, $\mathbf{q}_0$ and $\mathbf{p}_0$ are the initial coordinate and momentum, and $f_B(\mathbf{q}_0, \mathbf{p}_0)$ is the phase space distribution of the initial state. $C_{\text{cl}}(t)$ is a real function and $C_{\text{cl}}(t) = C_{\text{cl}}(-t)$, whereas the quantum bath correlation function $C(t)$ is a complex function and one can show that $C(t) = C^*(-t)$. Clearly,

$$C_{\text{cl}}(t) \neq C(t) \quad (28)$$

For the harmonic oscillator bath,

$$H_B(\mathbf{q}, \mathbf{p}) = \sum_j \frac{1}{2} p_j^2 + \frac{1}{2} \omega_j^2 q_j^2 \quad (29)$$

$$B(\mathbf{q}, \mathbf{p}) = - \sum_j \sqrt{\frac{2\omega_j}{\hbar}} \gamma_j q_j = - \sum_j (\omega_j^2 \delta q_j) q_j \quad (30)$$

Here we briefly discuss the physical meaning of the quantity $B(\mathbf{q}, \mathbf{p})$ in the context of photo-induced electron and energy transfer. In this case, the states $|m\rangle$ correspond to some electronic excited states. Let $|G\rangle$ denote the electronic ground state, $|G\rangle \notin \{|m\rangle\}$. The potential energy of the electronic state $|m\rangle$ at a given bath coordinate $\mathbf{q}$ is

$$E_m(\mathbf{q}) = \varepsilon_m + \lambda_m + \sum_j \frac{1}{2} \omega_j^2 q_j^2 - \sum_j (\omega_j^2 \delta q_{j,m}) q_j \quad (31)$$

whereas the potential energy of the ground state is

$$E_G(\mathbf{q}) = \varepsilon_G + \sum_j \frac{1}{2} \omega_j^2 q_j^2 \quad (32)$$

The vertical excitation energy of the electronic state $|m\rangle$ is thus

$$E_{\text{ex},m}(\mathbf{q}) = (\varepsilon_m - \varepsilon_G) + \lambda_m - \sum_j (\omega_j^2 \delta q_{j,m}) q_j \quad (33)$$

and the expectation of the vertical excitation energy is

$$\langle E_{\text{ex},m} \rangle_G = (\varepsilon_m - \varepsilon_G) + \lambda_m \quad (34)$$

because $\langle q_j \rangle_G = 0$. Therefore,

$$\boxed{B(\mathbf{q}, \mathbf{p}) = E_{\text{ex},m}(\mathbf{q}) - \langle E_{\text{ex},m} \rangle_G \equiv \delta E_{\text{ex},m}(\mathbf{q})} \quad (35)$$

In the context of photo-induced electron and energy transfer, the quantity $B(\mathbf{q}, \mathbf{p})$ corresponds to the fluctuation of the excitation energy due to bath vibration.

The solution of the classical equation of motion of bath mode j is

$$q_j(t) = q_{j,0} \cos(\omega_j t) + \frac{p_{j,0}}{\omega_j} \sin(\omega_j t) \quad (36)$$

Since the bath modes are independent, the distribution function can be factorized, $f(\mathbf{q}_0, \mathbf{p}_0) = \prod_j f(q_{j,0}, p_{j,0})$. For notational simplicity, we will drop the subscript 0 for initial coordinate and momentum from hereon. The classical bath correlation function becomes

$$\begin{aligned} C_{\text{cl}}(t) &= \frac{2}{\hbar} \sum_{j,k} \sqrt{\omega_j \omega_k} \gamma_j \gamma_k \prod_l \iint dq_l dp_l f(q_l, p_l) \left[q_j \cos(\omega_j t) + \frac{p_j}{\omega_j} \sin(\omega_j t) \right] q_k \\ &= \frac{2}{\hbar} \sum_j \omega_j \gamma_j^2 \iint dq_j dp_j f(q_j, p_j) \left[q_j \cos(\omega_j t) + \frac{p_j}{\omega_j} \sin(\omega_j t) \right] q_j \\ &\quad + \frac{2}{\hbar} \sum_{j \neq k} \sqrt{\omega_j \omega_k} \gamma_j \gamma_k \iint dq_j dp_j f(q_j, p_j) \left[q_j \cos(\omega_j t) + \frac{p_j}{\omega_j} \sin(\omega_j t) \right] \iint dq_k dp_k f(q_k, p_k) q_k \end{aligned} \quad (37)$$

In the last equality, the second term is zero because $\iint dq_k dp_k f(q_k, p_k) q_k = 0$. Therefore,

$$C_{\text{cl}}(t) = \frac{2}{\hbar} \sum_j \omega_j \gamma_j^2 \iint dq_j dp_j f(q_j, p_j) \left[q_j^2 \cos(\omega_j t) + \frac{p_j q_j}{\omega_j} \sin(\omega_j t) \right] \quad (38)$$

We will consider two different distribution functions for the bath, the Boltzmann distribution and the Wigner distribution, corresponding to the fully classical and semiclassical cases, respectively.

3.1 Boltzmann Distribution

The Boltzmann distribution function of the harmonic oscillator is

$$f_{\text{Boltz}}(q_j, p_j) = \frac{1}{Z} \exp \left[-\frac{\beta}{2} (p_j^2 + \omega_j^2 q_j^2) \right] \quad (39)$$

where Z is the partition function,

$$Z = \iint dq_j dp_j \exp \left[-\frac{\beta}{2} (p_j^2 + \omega_j^2 q_j^2) \right] = \frac{2\pi}{\beta\omega_j} \quad (40)$$

Using the properties of Gaussian functions, we can show that

$$\iint dq_j dp_j f_{\text{Boltz}}(q_j, p_j) p_j q_j = 0 \quad (41)$$

$$\iint dq_j dp_j f_{\text{Boltz}}(q_j, p_j) q_j^2 = \frac{1}{\beta\omega_j^2} \quad (42)$$

Therefore, the **classical bath correlation function with Boltzmann distribution** is

$$C_{\text{Boltz}}(t) = \frac{2}{\beta\hbar} \sum_j \frac{\gamma_j^2}{\omega_j} \cos(\omega_j t) \quad (43)$$

The Fourier transform of $C_{\text{Boltz}}(t)$ is

$$C_{\text{Boltz}}(\omega) = \frac{2\pi}{\beta\hbar} \sum_j \frac{\gamma_j^2}{\omega_j} [\delta(\omega + \omega_j) + \delta(\omega - \omega_j)] \quad (44)$$

or in terms of $J(\omega)$,

$$C_{\text{Boltz}}(\omega) = \frac{2\pi}{\beta\hbar\omega} [J(\omega) - J(-\omega)] \quad (45)$$

It's easy to show that $C_{\text{Boltz}}(\omega)$ is an even function of ω . Since $J(\omega) = 0$ when $\omega < 0$, we can solve for $J(\omega)$,

$$J(\omega) = \frac{\beta\hbar\omega}{2\pi} \Theta(\omega) C_{\text{Boltz}}(\omega) \quad (46)$$

where $\Theta(\omega)$ is the Heaviside step function. Using the relation $C_{\text{Boltz}}(t) = C_{\text{Boltz}}(-t)$, one can show that

$$C_{\text{Boltz}}(\omega) = 2 \int_0^\infty dt C_{\text{Boltz}}(t) \cos(\omega t) \quad (47)$$

Substituting this equation to the expression of $J(\omega)$, we obtain the commonly used form for the spectral density

$$J(\omega) = \frac{\beta\hbar\omega}{\pi} \int_0^\infty dt C_{\text{Boltz}}(t) \cos(\omega t) \quad (48)$$

This equation shows how to obtain the spectral density from a classical bath correlation function (excitation energy autocorrelation function) with Boltzmann distribution.

The Heaviside function is omitted in this expression.

3.2 Wigner Distribution

The Wigner distribution function of the harmonic oscillator is

$$f_{\text{Wig}}(q_j, p_j) = \frac{1}{\pi\hbar} \tanh\left(\frac{\beta\hbar\omega_j}{2}\right) \exp\left[-\tanh\left(\frac{\beta\hbar\omega_j}{2}\right) \left(\frac{1}{\hbar\omega_j}p_j^2 + \frac{\omega_j}{\hbar}q_j^2\right)\right] \quad (49)$$

Thus

$$\iint dq_j dp_j f_{\text{Wig}}(q_j, p_j) p_j q_j = 0 \quad (50)$$

$$\iint dq_j dp_j f_{\text{Wig}}(q_j, p_j) q_j^2 = \frac{\hbar}{2\omega_j} \coth\left(\frac{\beta\hbar\omega_j}{2}\right) \quad (51)$$

The **semiclassical bath correlation function with Wigner distribution** is

$$C_{\text{Wig}}(t) = \sum_j \gamma_j^2 \coth\left(\frac{\beta\hbar\omega_j}{2}\right) \cos(\omega_j t) \quad (52)$$

The Fourier transform of $C_{\text{Boltz}}(t)$ is

$$C_{\text{Wig}}(\omega) = \pi \sum_j \gamma_j^2 \coth\left(\frac{\beta\hbar\omega_j}{2}\right) [\delta(\omega + \omega_j) + \delta(\omega - \omega_j)] \quad (53)$$

or in terms of $J(\omega)$,

$$C_{\text{Boltz}}(\omega) = \pi \coth\left(\frac{\beta\hbar\omega}{2}\right) [J(\omega) - J(-\omega)] \quad (54)$$

Similarly, **one can obtain the spectral density from a semiclassical bath correlation function with Wigner distribution** by

$$\boxed{J(\omega) = \frac{2}{\pi} \tanh\left(\frac{\beta\hbar\omega}{2}\right) \int_0^\infty dt C_{\text{Wig}}(t) \cos(\omega t)} \quad (55)$$

Appendix A Wigner Representation of Density Operator

The Wigner transform maps the quantum mechanical density operator to a phase space distribution function i.e., the Wigner distribution. Consider a one-dimensional case, the coordinate representation of the density operator is

$$\rho(x, x', t) = \langle x | \hat{\rho}(t) | x' \rangle \quad (\text{A1})$$

We define the mid-point and difference of the coordinates x and x' as

$$q = \frac{x + x'}{2}, \quad s = x - x' \quad (\text{A2})$$

Then the coordinate representation of the density operator can be expressed in terms of q and s as

$$\rho(q, s, t) = \left\langle q + \frac{s}{2} \left| \hat{\rho}(t) \right| q - \frac{s}{2} \right\rangle \quad (\text{A3})$$

The **Wigner representation of the density operator** is defined as the Fourier transform of the coordinate representation with respect to s ,

$$\rho_{\text{Wig}}(q, p, t) = \int_{-\infty}^{\infty} ds e^{-\frac{i}{\hbar} ps} \left\langle q + \frac{s}{2} \left| \hat{\rho}(t) \right| q - \frac{s}{2} \right\rangle \quad (\text{A4})$$

And the **Wigner distribution function** is defined as

$$f_{\text{Wig}}(q, p, t) = \frac{1}{2\pi\hbar} \rho_{\text{Wig}}(q, p, t) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} ds e^{-\frac{i}{\hbar} ps} \left\langle q + \frac{s}{2} \left| \hat{\rho}(t) \right| q - \frac{s}{2} \right\rangle \quad (\text{A5})$$

For a pure state

$$\hat{\rho} = |\Psi\rangle\langle\Psi| \quad (\text{A6})$$

The Wigner representation is

$$\rho_{\text{Wig}}(q, p) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} ds e^{-\frac{i}{\hbar} ps} \Psi^*(q + s/2) \Psi(q - s/2) \quad (\text{A7})$$

One can show that

$$\int_{-\infty}^{\infty} dp \rho_{\text{Wig}}(q, p) = |\Psi(q)|^2 \quad (\text{A8})$$

$$\int_{-\infty}^{\infty} dq \rho_{\text{Wig}}(q, p) = |\tilde{\Psi}(p)|^2 \quad (\text{A9})$$

where $\tilde{\Psi}(p)$ is the momentum representation of the state $|\Psi\rangle$, and

$$\int_{-\infty}^{\infty} dq \int_{-\infty}^{\infty} dp \rho_{\text{Wig}}(q, p) = 1 \quad (\text{A10})$$

However, $\rho_{\text{Wig}}(q, p)$ can have negative values, so it is a quasiprobability distribution instead of a probability distribution.

For the density operator of harmonic oscillator at thermal equilibrium,

$$\hat{\rho} = \frac{e^{-\beta\hat{H}}}{\text{Tr}_{\text{B}} e^{-\beta\hat{H}}} \quad (\text{A11})$$

$$\hat{H} = \frac{\hat{p}^2}{2} + \frac{1}{2}\omega^2\hat{q}^2 \quad (\text{A12})$$

The coordinate representation of the density operator takes the form

$$\rho(q, s) = \left[\frac{\omega}{\pi\hbar} \tanh\left(\frac{\beta\hbar\omega}{2}\right) \right]^{1/2} \exp\left\{ -\frac{\omega}{\hbar} \left[\tanh\left(\frac{\beta\hbar\omega}{2}\right) q^2 + \frac{1}{4} \coth\left(\frac{\beta\hbar\omega}{2}\right) s^2 \right] \right\} \quad (\text{A13})$$

Perform the Fourier transform with respect to s , we have

$$f_{\text{Wig}}(q, p) = \frac{1}{\pi\hbar} \tanh\left(\frac{\beta\hbar\omega}{2}\right) \exp\left[-\tanh\left(\frac{\beta\hbar\omega}{2}\right) \left(\frac{\omega}{\hbar} q^2 + \frac{1}{\hbar\omega} p^2 \right) \right] \quad (\text{A14})$$

Finally, we express the quantum mechanical thermal expectation value as phase space integral using Wigner representation. The Wigner representation of an arbitrary operator $\hat{A}$ is

$$A_{\text{Wig}} = \int_{-\infty}^{\infty} ds e^{-\frac{i}{\hbar}ps} \left\langle q + \frac{s}{2} \left| \hat{A} \right| q - \frac{s}{2} \right\rangle \quad (\text{A15})$$

Note that $q_{\text{Wig}} = q$ and $p_{\text{Wig}} = p$. The quantum mechanical thermal expectation value in coordinate representation is

$$\begin{aligned} \text{Tr}(\hat{\rho}\hat{A}) &= \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dx' \langle x | \hat{\rho} | x' \rangle \langle x' | \hat{A} | x \rangle \\ &= \int_{-\infty}^{\infty} dq \int_{-\infty}^{\infty} ds \left\langle q + \frac{s}{2} \left| \hat{\rho} \right| q - \frac{s}{2} \right\rangle \left\langle q - \frac{s}{2} \left| \hat{A} \right| q + \frac{s}{2} \right\rangle \end{aligned} \quad (\text{A16})$$

where we have used the Jacobian determinate for the change of variables $(x, x') \rightarrow (q, s)$. We rewrite the integral as

$$\begin{aligned} \text{Tr}(\hat{\rho}\hat{A}) &= \int_{-\infty}^{\infty} dq \int_{-\infty}^{\infty} ds \int_{-\infty}^{\infty} ds' \delta(s' - s) \left\langle q + \frac{s}{2} \left| \hat{\rho} \right| q - \frac{s}{2} \right\rangle \left\langle q - \frac{s'}{2} \left| \hat{A} \right| q + \frac{s'}{2} \right\rangle \\ &= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dq \int_{-\infty}^{\infty} ds \int_{-\infty}^{\infty} ds' \int_{-\infty}^{\infty} dp e^{\frac{i}{\hbar}p(s'-s)} \left\langle q + \frac{s}{2} \left| \hat{\rho} \right| q - \frac{s}{2} \right\rangle \left\langle q - \frac{s'}{2} \left| \hat{A} \right| q + \frac{s'}{2} \right\rangle \\ &= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dq \int_{-\infty}^{\infty} dp \left(\int_{-\infty}^{\infty} ds e^{-\frac{i}{\hbar}ps} \left\langle q + \frac{s}{2} \left| \hat{\rho} \right| q - \frac{s}{2} \right\rangle \right) \\ &\quad \times \left(\int_{-\infty}^{\infty} ds' e^{\frac{i}{\hbar}ps'} \left\langle q - \frac{s'}{2} \left| \hat{A} \right| q + \frac{s'}{2} \right\rangle \right) \\ &= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dq \int_{-\infty}^{\infty} dp \left(\int_{-\infty}^{\infty} ds e^{-\frac{i}{\hbar}ps} \left\langle q + \frac{s}{2} \left| \hat{\rho} \right| q - \frac{s}{2} \right\rangle \right) \\ &\quad \times \left(\int_{-\infty}^{\infty} ds' e^{-\frac{i}{\hbar}ps'} \left\langle q + \frac{s'}{2} \left| \hat{A} \right| q - \frac{s'}{2} \right\rangle \right) \end{aligned} \quad (\text{A17})$$

Therefore, the quantum mechanical thermal expectation value can be expressed via the phase space integral as

$$\text{Tr}(\hat{\rho}\hat{A}) = \int_{-\infty}^{\infty} dq \int_{-\infty}^{\infty} dp f_{\text{Wig}}(q, p) A_{\text{Wig}}(q, p) \quad (\text{A18})$$